GAGE BALDASSARO

Portfolio | LinkedIn | gb68@rice.edu | (504) 494-7226
Junior Developer focused on Graphics and Gameplay Programming

EDUCATION

Bachelor of Science in Computer Science | GPA: 3.83/4.0 August 2024 - May 2028
Rice University, Houston, TX
Courses Linear Algebra, Algorithms, Machine Learning, Multivariable Calculus, Differential Equations, Probability and Statistics, Physics (Mechanics, E&M, Waves, Thermodynamics)

PROJECTS

OpenGL Renderer | Solo Project July - August 2026
C++, GLSL, OpenGL

- Engineered a custom multi-pass rendering pipeline from scratch using C++, OpenGL, and GLSL.
- Programmed complex Blinn-Phong lighting and material system using tangent-space matrices for accurate normal mapping.
- Developed configurable post-processing shaders and robust 3D model loading via Assimp to render detailed environments and objects.

Code Athens | Lead Programmer July 2026
C#, Unity | Top 1% of 10.5k Submissions to GMTK Game Jam 2026 (Narrative, Audio)

- Collaborated with 5 designers and artists to prototype and ship a polished game within 96 hours.
- Designed an event-driven dialogue engine and a modular, asynchronous loading pipeline for seamless level transitions.
- Rapidly prototyped and developed drag-and-drop minigames and refined based on team feedback and playtesting.

Mech Combat Demo | Solo Project May - July 2026
C#, Unity

- Prototyped 3rd-person mech combat inspired by *Armored Core VI*, with a finite-state-machine player movement controller for energy-limited dashing, hovering, and boosting.
- Built a custom lock-on camera system with occlusion-aware target searching and velocity-based tilt.
- Designed a ScriptableObject weapon system with quadratic-intercept lead targeting for moving enemies.

EXPERIENCE

Unity Developer Intern | *Planetary Parfait* October 2025 - August 2026
Experiential Pixels Lab at Rice ECE, Houston, TX

- Engineered procedural terrain integration, collaborating with cross-functional teams to seamlessly render 2D Lunar data parsed from *JMARS* JSON API onto 3D Geospatial Lunar data from *Cesium*.
- Shipped 20+ bug fixes and gameplay features based on community feedback and team playtesting to public Steam build.
- Coauthored scientific publication (forthcoming) on immersive visualization in geospatial scientific workflows.

Electrical Engineering Intern June - July 2025
Petrotech Inc., New Orleans, LA

- Designed and cleanly documented Modbus TCP/IP communication protocol enabling reliable PLC-to-HMI data exchange via DIGI One IAP.
- Programmed a Human-Machine Interface (HMI) in Excel VBA for monitoring the Programmable Logic Controller of an ALLISON 501-KB Gas Turbine.
- Developed HMI screens for monitoring PLC based on complex engineering drawings using FactoryTalk View Machine software.

America Reads Mentor September - December 2025
Nehemiah Center, Houston, TX

- Mentored 1st through 4th grade students in literacy and mathematics in afterschool program for students of disadvantaged families in the Houston area.

SKILLS

Programming Languages C#, C++, C, Python, Java, GLSL, HLSL, Lua, MATLAB
Technical Skills Unity, Git, GitHub, OpenGL, Assimp, Microsoft Office, JMARS, Cesium, SteamWorks